

Math 571 - Exam 1 (110 points)

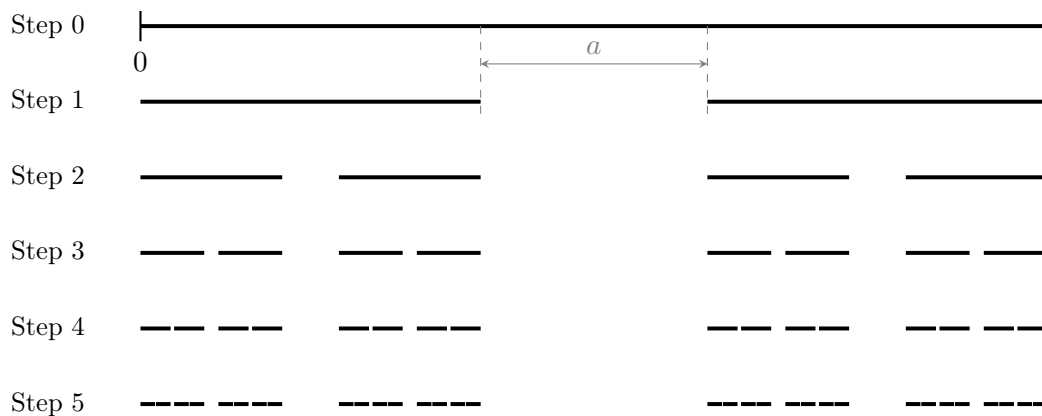
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Question 1 (15 points). Analyze the convergence or divergence of the following series. If the series is a power series, give the radius of convergence.

- (a) $\sum_{n=1}^{\infty} \frac{2+\sin(n)}{n}$
 (b) $\sum_{n=0}^{\infty} \frac{z^n}{(3+(-1)^n)^n}$

Question 2 (15 points). There is a new game where you and your opponent try to control a bounded region of space. Each ship you deploy can control a spherical region of a fixed radius called a *unit*. You are not allowed to place two of your own ships within one unit of each other. If you must place a ship within a region of space controlled by your opponent, then you must battle. The goal is to eventually control all of your region of space. Can you play so that battles never occur?

Question 3 (15 points). Consider a *fat Cantor set* formed by removing the middle a^{th} of the unit interval at each step, for $0 \leq a \leq \frac{1}{3}$. So you remove a , then $2a^2$, then $4a^3$, etc. The total amount removed is $a(1 + 2a + 2^2a^2 + \cdots) = a(\sum_{i=0}^{\infty} (2a)^i) = \frac{a}{1-2a}$ and so the measure of the a -Cantor set is $1 - \frac{a}{1-2a} = \frac{1-3a}{1-2a}$.



For each of the following properties, state whether the a -Cantor set (with its inherited topology) possesses it. Provide a single-line reason based on results in the text, citing the specific theorem or definition.

- Closed
- Open
- Metric
- Compact
- Perfect
- Uncountable
- Connected
- Complete

Question 4 (20 points). Show that every infinite sequence of real numbers has a monotone subsequence that converges to $\limsup_i x_i$.

Question 5 (20 points). Let (X, d) be a metric space. Let A and B be closed sets so that $A \cap B = \emptyset$. Show that there are open sets U and V so that $A \subseteq U$, $B \subseteq V$, and $U \cap V = \emptyset$.

Question 6 (20 points). Consider $X = \mathbb{R} \cup \{\infty\}$ let $A \subseteq X$ be open if

- A is an open subset of $\mathbb{R}$ or
- $\infty \in A$ and $X \setminus A$ is compact.

Show the following

- X is a topological space.
- X is compact.
- A metric can be placed on X that generates this topology. (There are actually many ways to do this. It might help to visualize the situations, and there are many ways to do this as well.)